

Vectors & The Geometry of Space

Distance between $P_1(x_1, y_1, z_1)$ & $P_2(x_2, y_2, z_2)$

$$|P_1 P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Standard equation for sphere radius a center (x_0, y_0, z_0)

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = a^2$$

Magnitude of vector $|V|$

$$|V| = \sqrt{V_1^2 + V_2^2 + V_3^2} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Angle between two vectors

$$\theta = \cos^{-1} \left(\frac{u_1 v_1 + u_2 v_2 + u_3 v_3}{|u| |v|} \right) \quad \cos \theta = \frac{u \cdot v}{|u| |v|}$$

Vector projection of u onto v :

$$\text{proj}_v u = \left(\frac{u \cdot v}{|v|^2} \right) v$$

* scalar component of u

$$|u| \cos \theta = \frac{u \cdot v}{|v|} = u \cdot \frac{v}{|v|}$$

Work by constant force

$$W = F \cdot D = |F| |D| \cos \theta$$

u — vector parallel to v + vector orthogonal to v

$$u = \text{proj}_v u + (u - \text{proj}_v u)$$

$$u = \left(\frac{u \cdot v}{|v|^2} \right) v + \left(u - \frac{u \cdot v}{|v|^2} v \right)$$

Cross product

$$u \times v = (|u||v| \sin \theta) n$$

n = unit vector

*parallel vectors $a \times b = 0$

Area of parallelogram

$$|u \times v| = |u||v| |\sin \theta| |n| = |u||v| \sin \theta$$

Calculation of cross product by using determinants

$$u = u_1 i + u_2 j + u_3 k$$

$$v = v_1 i + v_2 j + v_3 k$$

$$u \times v = \begin{vmatrix} i & j & k \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

Triple scalar product

$$(U \times V) \cdot W = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$$

Vector equation for a line

$$r(t) = r_0 + tv \quad -\infty < t < \infty \quad r_0 \quad P_0(x_0, y_0, z_0)$$

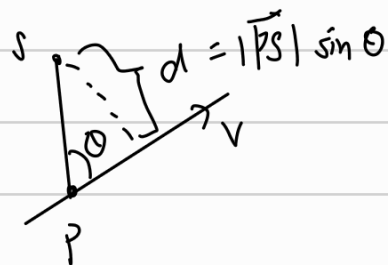
Parametric Equations for a line

$$v = v_1 i + v_2 j + v_3 k$$

$$x = x_0 + tv_1, \quad y = y_0 + tv_2, \quad z = z_0 + tv_3 \quad -\infty < t < \infty$$

Distance from a Point S to a line through P parallel to v

$$d = \frac{|\vec{PS} \times \vec{v}|}{|\vec{v}|}$$



Equation for a plane

Plane through $P_0(x_0, y_0, z_0)$ normal to $n = Ai + Bj + Ck$

Vector equations $n \cdot \vec{P_0P} = 0$

Component equation $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$

Distance from point to plane

$$d = \left| \vec{PS} \cdot \frac{\vec{n}}{|\vec{n}|} \right|$$

Quadratic Surfaces

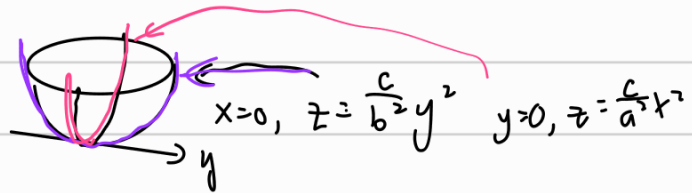


i) Ellipsoids

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad |x| \leq a, |y| \leq b, |z| \leq c$$

ii) Paraboloids

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c}$$



iii) Cones

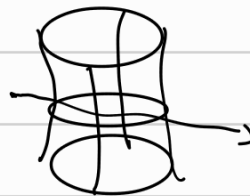
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$$



$$\text{line } z = \pm \frac{c}{b} y \quad z = \pm \frac{c}{a} x$$

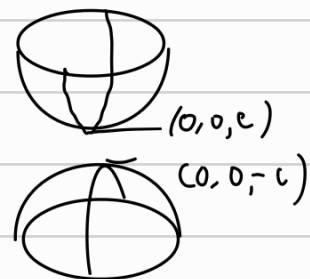
iv) Hyperboloids
of one sheet

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$



v) Hyperboloids
of two sheets

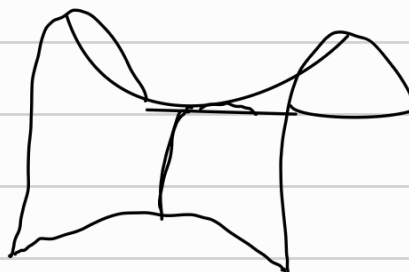
$$\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$



vi) Saddle point

Hyperbolic paraboloid

$$\frac{y^2}{b^2} - \frac{x^2}{a^2} = \frac{z}{c}, \quad c > 0$$



Chap 12 Prac

3) -2u a) component form b) magnitude

a) $\langle 6, -8 \rangle$ b) 10

4) 5v a) $\langle 10, -25 \rangle$ b) $5\sqrt{29}$

7) $\frac{8}{\sqrt{17}}\mathbf{i} - \frac{2}{\sqrt{17}}\mathbf{j}$ 8) $-3\mathbf{i} - 4\mathbf{j}$

11) $v(\frac{\pi}{2}) = -2\sin\frac{\pi}{2}\mathbf{i} + 2\cos\frac{\pi}{2}\mathbf{j} = -2\mathbf{i}$ length = 2 direction = $-\mathbf{i}$

12) $v(\ln 2) = (2\cos\ln 2 - 2\sin\ln 2)\mathbf{j} + (2\sin\ln 2 + 2\cos\ln 2)\mathbf{j}$
 length = $\sqrt{8} = 2\sqrt{2}$ direction = $\frac{\cos\ln 2 - \sin\ln 2}{\sqrt{2}}\mathbf{i} + \frac{\sin\ln 2 + \cos\ln 2}{\sqrt{2}}\mathbf{j}$

17) $V = \mathbf{i} + \mathbf{j}$ $|V| = \sqrt{2}$ $V \cdot u = 3 = u \cdot V$ $V \times u = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 0 \\ 2 & 1 & -2 \end{vmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} \\ 1 & 1 \\ 2 & 1 \end{vmatrix}$
 $u = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ $|u| = 3$
 $u \times v = -(v \times u) = 2\mathbf{i} - 2\mathbf{j} + \mathbf{k}$
 $|v \times u| = \sqrt{4+4+1} = 3$
 $= -2\mathbf{i} + \mathbf{k} - 2\mathbf{k} + 2\mathbf{j}$
 $= -2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$

$\theta = \cos^{-1}\left(\frac{u \cdot v}{|u||v|}\right) = \cos^{-1}\left(\frac{3}{3\sqrt{2}}\right) = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$
 scalar component of $u = |u| \cos \theta = \frac{u \cdot v}{|v|} = \frac{3}{\sqrt{2}}$

$$\text{proj}_v u = \left(\frac{u \cdot v}{|v|^2}\right)v = \frac{3}{2}(\mathbf{i} + \mathbf{j})$$

19) $v = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$
 $u = \mathbf{i} + \mathbf{j} - 5\mathbf{k}$

$$u = \text{proj}_v u + (u - \text{proj}_v u)$$

$$u = \frac{8}{6}(2\mathbf{i} + \mathbf{j} - \mathbf{k}) + \left[\mathbf{i} + \mathbf{j} - 5\mathbf{k} - \left(\frac{8}{3}\mathbf{i} + \frac{4}{3}\mathbf{j} - \frac{4}{3}\mathbf{k}\right)\right]$$

$$\left(\frac{u \cdot v}{|v|^2}\right)v$$

$$u = \frac{4}{3}(2\mathbf{i} + \mathbf{j} - \mathbf{k}) - \frac{1}{3}(5\mathbf{i} + \mathbf{j} + 11\mathbf{k})$$

23) $v = v_1 \hat{i} + v_2 \hat{j} + v_3 \hat{k}$ $w = w_1 \hat{i} + w_2 \hat{j} + w_3 \hat{k}$

$$\sqrt{V_1^2 + V_2^2 + V_3^2} = 2$$

$$\sqrt{w_1^2 + w_2^2 + w_3^2} = 3$$

$$\frac{1}{2} = \frac{uV}{|uV|} \quad \} = uV$$

$$V_1 W_1 + V_2 W_2 + V_3 W_3 = 3$$

$$\omega_1^2 + \omega_2^2 + \omega_3^2 = 0$$

$$V_1^2 + V_2^2 + V_3^2 = 4$$

$$|v - 2w| = \sqrt{(v_1 - 2w_1)^2 + (v_2 - 2w_2)^2 + (v_3 - 2w_3)^2}$$

$$= \sqrt{K^2 + 4W_1^2 + V_2^2 + 4W_2^2 + V_3^2 + 4W_3^2 - 4KW_1 - 4V_2W_2 - 4V_3W_3}$$

$$= \sqrt{4 \times 36 - 4(3)} = \sqrt{28} = 2\sqrt{7}$$

$$25) \quad (u \times v) = \begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ 2 & 1 & 1 \end{vmatrix} = \begin{vmatrix} i & j \\ 1 & 1 \\ 2 & 1 \end{vmatrix} = i(1-2) + j(1-2) + k(1-2) = -i - j - k = \sqrt{14}$$

$$(u \times v) \cdot w = \begin{vmatrix} 1 & 1 & -1 \\ 2 & 1 & 1 \\ 1 & -2 & 3 \end{vmatrix} = 3 - 1 + 4 - 1 + 2 - 6 = 1$$

$$2) \quad d = \frac{|\vec{p} \times \vec{r}|}{|\vec{v}|} \quad (2, 2, 1) \quad x = -t \quad y = t \quad z = 1 + t$$

passes through $P(0,0,-1)$ $v = -i + j + k$

$$\vec{PS} = (2, 2, 1)$$

$$|\vec{PS} \times \vec{r}| = \begin{vmatrix} i & j & k \\ 2 & 2 & 1 \\ -1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} i & j \\ 2 & 2 \\ -1 & -1 \end{vmatrix} = 2i - j + 2k + 2k - i - 2j = \sqrt{i - 3j + 4k} = \sqrt{18}$$

$$|v| = \sqrt{3}$$

$$d = \frac{\sqrt{26}}{\sqrt{3}} = \frac{\sqrt{78}}{3}$$

30) $d = \frac{|\vec{PS} \times \vec{v}|}{|\vec{v}|}$

$$1 \overrightarrow{RS} \times \overrightarrow{r} = \begin{vmatrix} 1 & j & k \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} i & j \\ -2 & 2 \\ 1 & 1 \end{vmatrix}$$

$$P(2, 2, 0)$$

$$S(0,4,1)$$

$$\vec{p}_S = (-2, 2, 1)$$

$$= i+j-2k-2k-i+j$$

$$r = i + j + k$$

$$d = \frac{5}{\sqrt{3}}$$

$$31) \quad x = 1 - 3t \quad y = 2 \quad z = 2 + 7t$$

$$32) \quad \vec{PQ} = (0, 1, -1) \quad x = 1, \quad y = 2 + t, \quad z = -t \quad 0 \leq t \leq 1$$

$$33) \quad S(6, 0, -6) \quad x - y = 4 \quad d = \left| \vec{PS} \cdot \frac{\vec{n}}{|\vec{n}|} \right|$$

$$P(4, 0, 0) \quad \vec{PS} = (2, 0, -6) \quad \vec{n} = i - j$$

$$\frac{2i - 6k \cdot i - j}{\sqrt{2}}$$

$$34) \quad S(3, 0, 10) \quad 2x + 3y + z = 2$$

$$P(1, 0, 0) \quad \vec{PS} = (2, 0, 10) \\ 2i + 10k$$

$$\vec{n} = 2i + 3j + k \quad d = \left| \frac{\vec{PS} \cdot \vec{n}}{|\vec{n}|} \right| = \frac{14}{\sqrt{14}} = \sqrt{14}$$

$$35) \quad 2(x-3) + (y+2) + z-1 = 0 \\ 2x + y + z = 5$$

$$36) \quad (x+1) - 2(y-6) + 3(z) = 0 \quad (-1, 6, 0) \quad \vec{n} = i - 2j + 3k \\ x - 2y + 3z = -13$$